Q1

for $\forall$ state i, j, $P_{ij}^n > 0$ $P_{ji}^m > 0$ for some n, m

define f_{ij} as the probability of passing through j in the process of going from i and returning to i

$f_{ij} \geq P_{ij}^n P_{ji}^m > 0$

$\therefore P(\exists n > 1 \text{ s.t. } X_n = j \mid X_0 = i) = \sum_{k=0}^{\infty} (1 - f_{ij})^k f_{ij} = 1$

Q2

a)

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ \vdots \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1-\lambda(1-\mu) & \lambda(1-\mu) & 0 & 0 & \cdots \\ (1-\lambda)\mu & 1-\lambda-\mu+2\lambda\mu & \lambda(1-\mu) & 0 & \cdots \\ 0 & (1-\lambda)\mu & 1-\lambda-\mu+2\lambda\mu & \lambda(1-\mu) & \\ \vdots & \vdots & \ddots & \ddots & \ddots \end{bmatrix} \end{array}$$

for $\forall$ state i, j, WLOG let $j > i$

$P_{ij}^{j-i} = [\lambda(1-\mu)]^{j-i} > 0$

$\therefore$ irreducible

b)

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ \vdots \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & \cdots \\ \frac{(1-\lambda)\mu}{\lambda+\mu-2\mu\lambda} & 0 & \frac{(1-\mu)\lambda}{\lambda+\mu-2\mu\lambda} & \\ 0 & \frac{(1-\lambda)\mu}{\lambda+\mu-2\mu\lambda} & 0 & \ddots \\ \vdots & \ddots & \ddots & \end{bmatrix} \end{array}$$

for $\forall i, j$, WLOG let $j > i$

$P_{ij}^{j-i} \geq [\lambda(1-\mu)]^{j-i} > 0$

$\therefore$ irreducible

$\lambda < \mu$

c)

let $p = \frac{(1-\mu)\lambda}{\lambda+\mu-2\mu\lambda}$, $q = \frac{(1-\lambda)\mu}{\lambda+\mu-2\mu\lambda}$

$f_0 = P\{\text{starting from 0 the chain will ever return to 0}\}$

$= 1 \cdot P\{\text{starting from 1 the chain will ever return to 0}\}$

$$= \begin{cases} \frac{(\frac{p}{q})^{N-1} - 1}{(\frac{p}{q})^{N} - 1} & \text{if } p \neq q \\ \frac{N-1}{N} & \text{if } p = q \end{cases}$$

where N is the capacity approaching to $+\infty$

$\therefore$ when $p \leq q$, i.e. $\lambda \leq \mu$, $f_0 \to 1$

when $p > q$, i.e. $\lambda > \mu$, $f_0 \to q/p = \frac{(1-\lambda)\mu}{(1-\mu)\lambda} < 1$

$\therefore$ the chain is $\begin{cases} \text{recurrent} & \text{if } \lambda \leq \mu \\ \text{transient} & \text{if } \lambda > \mu \end{cases}$

d)

in $\{X_n, n \geq 0\}$ the $P\{\text{starting from 1 the chain will ever return to 0}\} = 1$

if in $\{Y_m, m \geq 0\}$ the $P\{\text{starting from 1 the chain will ever return to 0}\} = 1$

$\therefore$ the chain is $\begin{cases} \text{recurrent} & \text{if } \lambda \leq \mu \\ \text{transient} & \text{if } \lambda > \mu \end{cases}$

e)

$\begin{cases} \underline{\Pi} = \underline{\Pi} P \\ \underline{\Pi} \mathbf{1} = 1 \end{cases}$

$\Rightarrow \pi_0 = \pi_0[1-(1-\mu)\lambda] + \pi_1(1-\lambda)\mu \Rightarrow \frac{\pi_1}{\pi_0} = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}$

$\pi_1 = \pi_0\lambda(1-\mu) + \pi_1(1-\mu-\lambda+2\mu\lambda) + \pi_2(1-\lambda)\mu \Rightarrow \frac{\pi_2}{\pi_1} = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}$

$\therefore \frac{\pi_{i+1}}{\pi_i} = t$, where $t = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}$

$\pi_0(1+t+t^2\cdots) = \pi_0\frac{1}{1-t} = 1$

$\Rightarrow \pi_0 = 1-t$

$\therefore \underline{\pi} = (1-t, (1-t)t, (1-t)t^2, \cdots (1-t)t^n, \cdots)$

Q3

a) $A =$ "ever inspecting i before return to 0"

let $q = 1-p$

for $p = 1/2$, $P(A) = p\cdot\frac{1}{i} + q\cdot\frac{1}{n+1-i} = \frac{1}{2}\left(\frac{1}{i}+\frac{1}{n+1-i}\right)$

for $p \neq 1/2, 0, 1$, $P(A) = p\cdot\frac{\frac{q}{p}-1}{(\frac{q}{p})^i-1} + q\cdot\frac{\frac{p}{q}-1}{(\frac{p}{q})^{n+1-i}-1}$

for $p = 0$ or 1, $P(A) = 1$

b) $B =$ "inspecting all states at least once before return to 0"

for $p = 1/2$, $P(B) = p\cdot\frac{1}{n} + q\cdot\frac{1}{n} = \frac{1}{n}$

for $p \neq 1/2, 0, 1$, $P(B) = p\cdot\frac{\frac{q}{p}-1}{(\frac{q}{p})^n-1} + q\cdot\frac{\frac{p}{q}-1}{(\frac{p}{q})^n-1}$

for $p = 0$ or 1, $P(B) = 1$

c) $C =$ "$T_1 < T_2 < \cdots < T_n$"

for $p = 1/2$, $P(C) = p = 1/2$

for $p = 0$, $P(C) = 0$

for $p = 1$, $P(C) = 1$

D. $\frac{q}{p}-1$

for $p \neq 1/2, 0, 1$ $\quad P(C) = \dfrac{1}{p \cdot \dfrac{q/p - 1}{(q/p)^n - 1} + q \cdot \dfrac{p/q - 1}{(p/q)^n - 1}} = \dfrac{1}{1 + (q/p)^n}$

Q4

$$E\left(\sum_{i=0}^{\infty} X_i\right) = \sum_{i=0}^{\infty} EX_i = \sum_{i=0}^{\infty} n \cdot \mu^i = n \cdot \frac{1}{1-\mu}$$